ORIGINAL RESEARCH ARTICLE

Perceptions, Feasibility, and Implementation Readiness of an Ambient Clinical AI Platform in Indian Outpatient Department Settings: A Pre- and Post-Demonstration Survey Study

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ABSTRACT

Background

India's healthcare system faces a chronic documentation burden exacerbated by a doctor-to-population ratio estimated at 0.6–0.9 per 1,000 population, with outpatient departments (OPDs) in high-volume specialties managing more than 200 patients per working day. Ambient clinical artificial intelligence (AI) platforms, which use automatic speech recognition (ASR), natural language processing (NLP), and large language models (LLMs) to generate structured clinical notes in real time, offer a potentially transformative solution. However, their feasibility within the uniquely complex linguistic and operational context of Indian clinical practice remains poorly characterised.

Objectives

To characterise the perceptions, concerns, and implementation readiness of a sample of Indian clinicians and medical affairs professionals towards the Mendalia clinical AI platform, measured before and after a structured live demonstration, and to contextualise these findings within the broader global evidence base for ambient AI scribes.

Methods

A prospective, pre–post survey study was conducted with 23 respondents (pre-demonstration) and 21 respondents (post-demonstration), drawn from Lupin Limited's Medical Affairs network across India. Surveys were administered via structured questionnaires covering documentation burden, AI familiarity, feature perceptions, workflow integration concerns, and adoption intent. Descriptive statistics were computed for each survey item. Qualitative feedback from the Medical Affairs team lead was also analysed.

Results

Pre-demonstration, 91.3% of respondents rated documentation burden as 4 or 5 on a 5-point scale, and 78.3% cited all of: missing historical data, patient forgetfulness, and lack of review time as compounding follow-up visit challenges. Post-demonstration, 76.2% of respondents rated the AI Scribing feature as capturing doctor–patient conversations either 'exceptionally well' or 'adequately'; 76.2% expressed high confidence in the platform's Clinical Reasoning output; 71.4% rated the Patient Notes and Visit Summarisation feature as 'critical' for chronic disease management; and 65% identified Ambient Scribing as the platform's strongest value proposition. The dominant concern was multilingual coverage, with 81% endorsing the need for regional Indian language support. Doctors with high patient footfalls in OPD were identified by 61.9% as the most appropriate target profile. The final adoption recommendation was strikingly positive: 85.7% of respondents recommended either an immediate or conditional pilot.

Conclusions

This study demonstrates strong pre-existing clinical need and high post-demonstration readiness for ambient AI scribing in Indian OPD settings. Multilingual NLP, data privacy compliance (specifically with India's Digital Personal Data Protection Act 2023), and workflow integration at scale remain the principal implementation

challenges. A structured, KOL-led pilot programme with rigorous real-world outcome evaluation is recommended as the immediate next step.

1. INTRODUCTION

India carries a paradoxical burden in its healthcare system: despite producing the largest number of medical graduates globally, the effective doctor-to-population ratio remains critically low. Accounting for attrition, migration, and the uneven rural, urban distribution of the physician workforce, the National Health Profile estimates that India's functional ratio stands between 0.6 and 0.9 doctors per 1,000 population, falling below the World Health Organization (WHO) benchmark of 1:1,000 in 17 of 28 states [1, 2]. This systemic scarcity is compounded by the extraordinary volume of patients seen in government and private OPDs alike. Studies conducted in tertiary care institutions have documented average physician consultation times of as little as 2.2 minutes per patient in high-volume specialties such as gastroenterology, neurology, and oncology, with individual OPDs registering more than 200 patient visits per working day [3].

The consequence is a physician workforce characterised by extraordinary patient throughput but inadequate time for meaningful clinical documentation. India accounts for approximately 17% of the global population and 20% of the global disease burden, yet 74% of its practising doctors reside in urban areas that serve only 28% of the population [2]. In this setting, high-quality contemporaneous clinical notes, essential for continuity of care, medicolegal protection, and chronic disease management, are routinely abbreviated, deferred, or omitted entirely.

Globally, artificial intelligence tools designed to reduce documentation burden have moved rapidly from research settings into clinical practice. The market for AI-powered clinical documentation was valued at approximately USD 2.5 billion in 2024 and is projected to reach USD 6.6 billion by 2031, growing at a compound annual growth rate exceeding 14% [4]. Among the most clinically relevant categories are ambient AI scribes, platforms that passively record and transcribe the physician–patient encounter, subsequently generating structured clinical notes using automatic speech recognition (ASR) and natural language processing (NLP). These tools have demonstrated documentation time reductions of 25–50% and physician burnout reductions of up to 13.9 percentage points in randomised controlled trials conducted in Western healthcare systems [5, 6].

India presents a uniquely complex target environment for such technology. With twenty-two constitutionally recognised languages, hundreds of regional dialects, and a deeply ingrained clinical culture in which code-switching between English and vernacular languages (particularly Hindi, Marathi, Tamil, Telugu, and Bengali) is routine during consultations, the NLP demands placed on any ambient AI system are considerably greater than those in predominantly English-speaking health systems. Additionally, the regulatory landscape is evolving: India's Digital Personal Data Protection Act (DPDP) of 2023 introduces new obligations around patient data consent, storage, and cross-border transfer that any deployed health AI tool must address.

Against this backdrop, the Mendalia AI platform, an ambient clinical AI system developed in Australia and designed for multilingual, high-volume clinical environments, was introduced to a cohort of Indian medical affairs professionals and clinicians affiliated with Lupin Limited. This paper reports the findings of a structured

pre- and post-demonstration survey study designed to characterise baseline perceptions of documentation burden and AI familiarity, assess the impact of a live demonstration on confidence and adoption intent, and identify the key facilitators and barriers to implementation in Indian OPD settings.

2. BACKGROUND AND LITERATURE CONTEXT

2.1 The Global Evidence Base for Ambient AI Scribes

The clinical utility of ambient AI documentation tools has been examined in an expanding body of peer-reviewed literature, predominantly from North American and European healthcare systems. A 2024 systematic review published in Perspectives in Health Information Management, drawing on 129 studies extracted from 673 candidate papers across PubMed, Embase, Scopus, and Web of Science, found that AI tools improved clinical documentation by structuring data, annotating notes, detecting errors, and serving as real-time documentation assistants, with 77% of studies focused on direct clinician benefit rather than coding or billing optimisation [7].

A landmark pragmatic randomised clinical trial published in 2025 compared two ambient AI scribe applications, Microsoft Dragon Ambient eXperience (DAX) Copilot and Nabla, against usual care across 238 outpatient physicians representing 14 specialties. Enrolled over a 60-day period (November 2024–January 2025), the trial demonstrated statistically significant reductions in documentation time-in-note as well as improvements in burnout scores measured by the Mini-Z 2.0 and the Professional Fulfillment Index, Work Exhaustion subscale [5]. These findings corroborate a JAMA Network Open study of 263 clinicians across six health systems, in which burnout prevalence dropped from 51.9% to 38.8% within 30 days of ambient AI scribe adoption [6].

A rapid review examining real-world evidence for digital scribes using ambient listening and generative AI in clinical practice (2014–2024), conducted according to the QUEST human evaluation framework and the SEIPS 3.0 model, highlighted consistent benefits in workflow efficiency and clinician satisfaction but also identified persistent barriers including multilingual support limitations, lack of direct EMR integration, and variability in accuracy across ASR and NLP components [8]. The reviewers noted explicitly that 'no published studies have evaluated clinical ambient scribe outcomes in multilingual clinical environments,' identifying this as a critical gap given the global prevalence of linguistically heterogeneous healthcare settings [9].

A prospective study conducted at Singapore General Hospital between December 2024 and May 2025, one of the first in an Asian multilingual setting, used standardised time-motion methodology across 169 consultations and demonstrated that an in-house ambient AI scribe ('Note Buddy'), capable of processing mixed-language transcripts using GPT-4o's native multilingual capabilities, significantly reduced documentation time while increasing eye contact during consultations [9]. This study is highly relevant to the Indian context given Singapore's multilingual clinical environment, which, while smaller in linguistic diversity, parallels India's code-switching challenges.

Performance evaluation of AI scribes remains methodologically heterogeneous. A scoping review of evaluation frameworks for ambient AI scribes (2020–2025), conducted according to PRISMA guidelines across PubMed, IEEE Explore, Scopus, Web of Science, and Embase, found that only seven studies met

rigorous inclusion criteria, and that evaluation approaches ranged from established NLP metrics (ROUGE, BERTScore) to novel human rating rubrics, with no standardised benchmark in use [10]. This absence of a gold-standard evaluation framework constitutes a major impediment to cross-system comparison and regulatory submission.

2.2 The Indian Healthcare Context

India's OPD system operates under structural constraints that have no close parallel in high-income countries. A cross-sectional study of nine super-specialty clinics at a government tertiary care hospital found that total OPD visits in a single year across these nine specialties exceeded 359,099, with gastroenterology, neurology, and oncology physicians seeing more than 30 patients per hour [3]. The study concluded that super-specialty clinics in government hospitals are 'overworked and understaffed,' and proposed structural interventions including parallel OPD streams, interventions that remain largely unimplemented due to workforce constraints.

This operational pressure translates directly into documentation quality deficits. The pre-demonstration survey data from the present study confirm that 91.3% of respondents rated the documentation burden faced by the Indian doctors they engage with as a 4 or 5 on a 5-point scale. Moreover, 78.3% identified the compound challenge of missing historical data, patient forgetfulness about previous prescriptions, and insufficient time to review prior notes as the dominant difficulty during follow-up visits, a finding with direct implications for chronic disease management.

India's non-communicable disease (NCD) burden adds further urgency. The country faces an estimated 101.3 million individuals living with diabetes and 315 million with hypertension, with chronic disease management requiring longitudinal documentation to support evidence-based titration and monitoring protocols [11]. The National Health Mission and National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Disease, and Stroke (NPCDCS) have both emphasised structured patient records as a prerequisite for effective chronic care delivery, yet the tools to generate such records at OPD scale remain almost entirely absent in routine Indian clinical practice.

2.3 Regulatory and Linguistic Considerations Specific to India

The Digital Personal Data Protection Act (DPDP) 2023 represents India's first comprehensive data privacy legislation, analogous in scope to the European Union's General Data Protection Regulation (GDPR). For clinical AI platforms, the DPDP introduces requirements for explicit patient consent prior to processing sensitive personal health data, restrictions on cross-border data transfer to countries not listed on an approved schedule, and obligations for data fiduciaries to implement proportionate security safeguards [12]. Any ambient AI scribe deployed in Indian OPD settings must articulate a DPDP-compliant data flow architecture, including data residency provisions, before clinical adoption.

India's linguistic complexity is unparalleled in scale. Among the 22 Eighth Schedule languages, Hindi (spoken by approximately 52.8% of the population), Bengali (8.0%), Marathi (7.0%), Telugu (6.9%), Tamil (5.7%), Gujarati (4.6%), Urdu (4.2%), and Kannada (3.6%) together account for the overwhelming majority of clinical consultations in their respective regions [13]. Beyond scheduled languages, regional dialects, Bhojpuri,

Maithili, Rajasthani, Bundeli, Chhattisgarhi, and many others, add further layers of phonological and lexical variation that standard ASR models trained on formal language corpora handle poorly. The present survey found that 81% of post-demonstration respondents endorsed the need for regional Indian language capture, identifying multilingual support as the platform's most critical development priority.

3. METHODS

3.1 Study Design

This study employed a prospective, pre–post survey design. Survey instruments were administered in two phases: before a live product demonstration of the Mendalia clinical AI platform, and after the demonstration. The design follows published frameworks for technology acceptance and readiness assessment in healthcare settings, including elements consistent with the Technology Acceptance Model (TAM) and the Systems Engineering Initiative for Patient Safety (SEIPS) 3.0 framework [8].

3.2 Participants

Survey respondents were medical affairs professionals, clinical liaison officers, and practising clinicians affiliated with Lupin Limited's Medical Affairs division across India. Respondents were recruited via the Medical Affairs team's internal network. Participation was voluntary and anonymous. Twenty-three individuals completed the pre-demonstration survey and 21 completed the post-demonstration survey. The difference in sample size reflects the voluntary, asynchronous nature of survey completion.

3.3 Survey Instruments

The pre-demonstration survey comprised eight items covering: (1) perceived documentation burden among engaged clinicians (rated 1–5); (2) most valued feature of AI medical scribing for Indian clinicians; (3) primary concern regarding AI scribing adoption in Indian OPD settings; (4) familiarity with AI clinical reasoning; (5) preferred role of AI in supporting clinical reasoning; (6) familiarity with AI-generated care plans; (7) perceived potential of AI-generated care plans to improve therapy adherence; and (8) most significant challenge during follow-up visits. Two additional items addressed the success metric and target doctor profile for a clinical AI tool.

The post-demonstration survey comprised nine items covering: (1) accuracy of AI Scribing in capturing doctor–patient conversations; (2) confidence in the Clinical Reasoning feature; (3) effectiveness of the Care Plan generation feature; (4) value of Patient Notes and Visit Summarisation for chronic disease management; (5) User Experience and workflow integration; (6) strongest value proposition/USP; (7) need for local Indian language capture; (8) most appropriate doctor profile target; (9) confidence in the platform's ability to maintain documentation quality and professionalism; and (10) final adoption recommendation.

3.4 The Mendalia Platform

The Mendalia AI platform is an ambient clinical intelligence system comprising four integrated modules: (a) AI Scribing, which uses real-time ASR to transcribe physician–patient encounters; (b) Clinical Reasoning, which generates differential diagnoses and evidence-based management suggestions aligned with international and Indian guidelines; (c) Care Plan Generation, which produces patient-facing and clinician-facing management plans; and (d) Patient Notes and Visit Summarisation, which produces longitudinal SOAP-format clinical notes and consultation summaries for chronic disease follow-up.

3.5 Demonstration Protocol

The live demonstration was conducted by Mendalia AI representatives and exposed participants to real-time operation of all four platform modules using simulated patient cases reflective of high-prevalence Indian clinical scenarios (type 2 diabetes mellitus, hypertension, and respiratory infection). Hindi-medium consultations were included to demonstrate multilingual ASR capability.

3.6 Statistical Analysis

Data were analysed descriptively. Proportions (%) were calculated for each response option within each survey item. Pre- and post-demonstration responses were compared narratively, with pre-demonstration data serving as a baseline characterisation of need and post-demonstration data interpreted as a measure of feature-specific acceptability following direct exposure. Qualitative thematic analysis was performed on the expert written feedback received from the Medical Affairs team lead (Dr. Kundan Nivangune), following established thematic coding procedures.

3.7 Ethics

The study involved survey administration to voluntary, anonymous respondents in an occupational context. No patient data were collected. No patient contact occurred during the research. The study was conducted in accordance with the principles of the Declaration of Helsinki (2013 revision). All participants provided implied informed consent through voluntary completion of the anonymous questionnaire.

4. RESULTS

4.1 Pre-Demonstration Findings: Characterising the Baseline Clinical Need

4.1.1 Documentation Burden

Documentation burden was rated highly by the respondent cohort. No respondents selected a burden score of 1 (no burden). Scores of 4 and 5 (high and very high burden) were reported by 60.9% and 30.4% of respondents respectively, yielding a combined high-burden prevalence of 91.3%. These data confirm that the population of clinicians engaged by Lupin's Medical Affairs network experiences documentation burden as a major operational challenge.

4.1.2 Most Valued AI Scribing Feature

When asked which aspect of AI Medical Scribing would provide the most immediate value to Indian clinicians, respondents divided across four options. Real-time ambient voice-to-text conversion was selected by 39.1%, followed by integration of scribed notes directly into existing EMR/EHR systems (34.8%), automatic structuring of unstructured notes into SOAP format (21.7%), and multi-language support (4.3%). The dominance of voice-to-text and EMR integration as priorities reflects the immediate operational bottlenecks in Indian clinical practice, where manual data entry represents the primary time cost.

4.1.3 Primary Adoption Concerns

The dominant pre-demonstration concern regarding AI scribing adoption in Indian OPD settings was accuracy of medical terminology in an Indian accent, selected by 65.2% of respondents. Patient privacy and data security was the second concern at 21.7%, followed by resistance from doctors to workflow change at 8.7%. Notably, high ambient noise in busy clinics, a factor that might be expected to rank more prominently given the well-documented acoustic complexity of Indian OPD environments, was the least selected option.

4.1.4 Familiarity with AI Clinical Reasoning

Prior familiarity with AI's ability to provide Clinical Reasoning support (differential diagnosis) was heterogeneous. A total of 47.8% described themselves as 'moderately familiar,' 34.8% as having 'heard of it but not understanding the mechanics,' 8.7% as 'not at all familiar,' and 8.7% as 'expert' (i.e., following the latest peer-reviewed studies on clinical AI). These figures indicate that the majority of respondents had a conceptual but non-technical awareness of AI clinical reasoning, a profile well-suited to a technology demonstration designed to bridge this gap.

4.1.5 AI and Follow-Up Visit Challenges

The question regarding the most significant challenge during follow-up visits produced a striking result: 78.3% of respondents selected 'All of the above,' indicating that missing historical data, patient forgetfulness about prescriptions and reports, and lack of time to review prior notes are experienced as simultaneous and compounding challenges rather than discrete issues. This aggregate burden directly maps onto the core value proposition of the Mendalia platform's longitudinal summarisation feature.

4.1.6 Success Metrics and Target Doctor Profile

When asked to identify the single most important metric for evaluating the success of a clinical AI platform, 47.8% selected 'Reduction in time spent per patient file,' followed by 'Accuracy of the clinical output' (34.8%), 'Ease of use and User Interface' (13%), and 'Patient satisfaction scores' (4.3%). Doctors with high patient footfalls in OPD were identified as the primary target group by 60.9% of respondents, followed by young doctors with 0–5 years of clinical practice (21.7%).

4.2 Post-Demonstration Findings: Feature-Specific Assessments

4.2.1 AI Scribing Accuracy

Following the live demonstration, 76.2% of respondents rated the AI Scribing feature as capturing the doctor–patient discussion either 'exceptionally well, requiring minimal to no editing' (23.8%) or 'adequately, capturing the core but needing minor technical corrections.' No respondents selected the 'poorly; requires significant manual rewriting' option. This high accuracy rating is particularly notable given that 65.2% of respondents had identified Indian accent recognition as their primary pre-demonstration concern, suggesting that the demonstration successfully addressed this concern.

4.2.2 Clinical Reasoning

The Clinical Reasoning feature (differential diagnosis and evidence-based suggestions) received a strong post-demonstration endorsement. A total of 76.2% of respondents reported 'high confidence: the logic was sound and aligned with global/Indian guidelines,' while 19% expressed 'cautious confidence: suggestions were good but felt too safe or generic.' No respondents selected 'no confidence' or 'low confidence noting clinical inaccuracies.' These findings are consistent with global evidence suggesting that, when adequately prompted and validated, LLM-based clinical reasoning can match or approach the quality of human expert differential diagnosis for a range of common presentations.

4.2.3 Care Plan Generation

Responses regarding the Care Plan generation feature were more distributed, reflecting the contextual complexity of Indian prescribing culture. The 'highly effective: plans are patient-friendly and actionable' option was selected by 28.6%, 'somewhat effective: good structure but may be too complex for some patients' by 38.1%, 'neutral: does not offer much more than a standard prescription' by 23.8%, and 'not effective: lacks the personalised human touch needed for adherence' by 9.5%. The written expert feedback from Dr. Nivangune acknowledged this limitation explicitly, noting that treatment plans 'are not that comprehensive, especially from an Indian practice perspective,' and observing that prescribing preferences for specific molecules and brands remain a domain of physician autonomy that AI tools appropriately should not supplant.

4.2.4 Patient Notes and Visit Summarisation

The Patient Notes and Visit Summarisation feature for chronic disease management received the most uniformly positive assessment of any post-demonstration item. A combined 95.2% of respondents rated this feature as either 'critical: this is the most valuable feature for long-term patient care' (71.4%) or 'useful: helps a busy doctor catch up quickly' (23.8%). This aligns with the pre-demonstration finding that 78.3% of respondents identified the compound challenge of missing historical data and follow-up preparation as their primary OPD concern.

4.2.5 User Experience and Workflow Integration

User experience perceptions were broadly positive. The largest group of respondents (61.9%) described the platform as 'seamless: it would easily fit into a standard 5–10-minute Indian OPD consultation.' A further 28.6% characterised it as 'manageable: requires some habit-changing but provides enough value to justify it.'

Only 9.5% felt the platform 'adds too many clicks/steps to the doctor's current workflow.' No respondents selected the 'impractical: not built for the high-volume environment of Indian healthcare' option.

4.2.6 Key Value Proposition

The platform's strongest value proposition was identified as 'Ambient Scribing (Time-saving)' by 65% of respondents, followed by 'Clinical Reasoning Support (Quality of care)' (20%), 'Care Plan Standardisation (As per latest guidelines)' (10%), and 'Longitudinal Follow-up Summaries (Patient retention)' (5%). This response profile reinforces that the immediate, high-value use case for Indian clinicians is time recovery through automated documentation rather than decision support, although Clinical Reasoning ranked as a meaningful secondary benefit.

4.2.7 Multilingual Requirements

The question of whether the platform should capture all local Indian languages during scribing produced a near-consensus response: 81% selected 'Yes,' with the remaining 19% selecting 'No.' No respondents selected 'Maybe.' This represents the strongest directional signal in the entire post-demonstration survey, underscoring that multilingual ASR is not merely a desirable feature enhancement but a prerequisite for viable deployment across India's linguistic geography.

4.2.8 Target Doctor Profile

Post-demonstration, respondents recalibrated their view of the optimal target doctor profile when compared with pre-demonstration expectations. Doctors with high patient footfalls in OPD remained the leading target group at 61.9%, but 'young tech-savvy doctors' moved up to 19% and 'Key Opinion Leaders (KOLs)' to 9.5%, with 'chronic therapies' specialists and 'doctors practising in metro cities' each receiving 9.5% of responses.

4.2.9 Overall Adoption Recommendation

The final question, 'What is your recommendation regarding the adoption/piloting of this platform?', produced a strongly favourable result. A combined 85.7% of respondents endorsed either 'Immediate Pilot: we should move to a small-scale trial with select KOLs' (52.4%) or 'Conditional Pilot: needs specific feature updates' (33.3%). The remaining 14.3% favoured 'Consider other clinical AI platforms or re-evaluate after 6–12 months.' Notably, no respondents selected 'Discard: not the right fit for our current medical strategy.'

5. DISCUSSION

5.1 The Alignment Between Survey Findings and Structural Healthcare Need

The pre-demonstration survey data presented in this study provide some of the first structured, quantitative characterisation of documentation burden as perceived by Indian medical affairs professionals and clinicians. The finding that 91.3% of respondents rated documentation burden at 4 or 5 on a 5-point scale is consistent with the structural parameters of Indian OPD practice documented in the literature. Pandey et al.'s cross-sectional study [3] found that high-volume OPD physicians averaged only 2.2 minutes per patient consultation, a figure that leaves virtually no time for contemporaneous documentation of the quality required

for continuity of care. In this context, the enormous enthusiasm for a platform promising to automate documentation is entirely rational.

The identification of follow-up visit challenges, specifically the compound of missing historical data, patient forgetfulness, and insufficient review time, by 78.3% of respondents as a singular aggregated problem maps precisely onto the visit summarisation and longitudinal note functions of the Mendalia platform. This is not merely a documentation problem; it is a patient safety issue. India's expanding NCD burden [11] requires chronic disease management protocols that depend on accurate, accessible longitudinal records. The near-universal enthusiasm for the Visit Summarisation feature (95.2% rating it 'critical' or 'useful') reflects a genuine unmet clinical need that AI tools are technically positioned to address.

5.2 Performance Ratings in Context

The post-demonstration scribing accuracy rating of 76.2% (exceptional or adequate) is notable when contextualised against the pre-demonstration concern about Indian accent recognition (65.2% flagged this as their primary concern). A demonstration that largely neutralised the most commonly anticipated barrier represents a significant first step toward technology acceptance. However, it must be acknowledged that a single demonstration event with simulated Hindi-medium cases represents a constrained test environment. Real-world performance in noisy, high-velocity OPD settings, with ambient conversations, interruptions, simultaneous attendants, and rapid dialect code-switching, will require prospective real-world evaluation.

The high confidence rating for Clinical Reasoning (76.2% expressing high confidence) is consistent with emerging global evidence. A 2024 study published in Nature Medicine found that adapted LLMs could outperform medical experts in clinical text summarisation across specialties [14]. However, the caveat raised by 19% of post-demonstration respondents, that Clinical Reasoning suggestions felt 'too safe or generic', aligns with known limitations of guideline-aligned LLMs. In the Indian context, where there is significant off-label prescribing, polypharmacy, and context-specific clinical pragmatism, AI reasoning outputs that adhere strictly to international guidelines may appear conservative relative to local clinical norms.

5.3 Multilingual NLP as a Non-Negotiable Implementation Requirement

The near-unanimous demand for regional Indian language support (81% endorsing 'Yes') elevates multilingual NLP from a product roadmap item to a clinical implementation prerequisite. This finding has deep implications for the architecture of any clinical AI system intended for Indian deployment. State-of-the-art multilingual ASR systems have advanced considerably: large transformer-based models such as OpenAI's Whisper and Google's Universal Speech Model (USM) now support recognition of dozens of languages, including several Indian scheduled languages. However, as demonstrated by the Singapore multilingual ambient scribe study [9], token-level language identification and LLM-based mixed-language summarisation remain active engineering challenges.

Beyond formal language coverage, Indian clinical encounters involve extensive phonological and lexical variation within languages. The pre-demonstration concern about 'specific Indian accents and dialects' and Dr. Nivangune's written note that 'many regional languages are spoken with some local variations' are consistent with the NLP literature, which recognises that model performance degrades substantially when tested on accented speech outside the training distribution. The practical recommendation, and a requirement before full commercial deployment, is that Mendalia AI's ASR components undergo structured performance evaluation across at minimum the eight most prevalent Indian languages, including code-switched Hindi–English conditions representative of urban OPD practice.

5.4 Data Privacy and Regulatory Compliance

The concerns raised regarding DPDP Act compliance, international regulatory approvals (TGA, CE-MDR, US FDA), privacy-preserving mechanisms, and professional body endorsement reflect a sophisticated and commercially appropriate set of due diligence queries from the Lupin Medical Affairs team. These concerns are not peripheral: they are likely to constitute the primary institutional gatekeeping criteria for any healthcare organisation or pharmaceutical company considering a formal AI platform deployment.

The DPDP Act 2023 introduces specific obligations for 'data fiduciaries' processing sensitive personal data, which unambiguously includes clinical consultation recordings and extracted health information. A DPDP-compliant deployment architecture must include: (a) explicit, disaggregated patient consent at the point of data collection; (b) clarity regarding whether audio recordings are stored or processed transiently; (c) specification of data residency (Indian servers or adequately protected international destinations); and (d) a published data retention and deletion policy. The question raised by Dr. Nivangune regarding 'SAHI and BODH', both Government of India privacy-preserving frameworks related to the Ayushman Bharat Digital Mission (ABDM), indicates awareness of the emerging Indian digital health ecosystem within which any deployed AI tool must operate.

5.5 Care Plan Limitations and the Prescribing Autonomy Consideration

The mixed response to the Care Plan generation feature (66.7% rating it 'somewhat effective' or 'neutral' or 'not effective') warrants careful interpretation. The written expert feedback from Dr. Nivangune provides important contextual nuance: treatment plans generated by the AI may be clinically sound but not 'comprehensive from an Indian practice perspective,' partly because the platform cannot incorporate physician brand preferences or molecule-level prescribing patterns. This observation reflects a fundamental principle in the deployment of clinical AI in pharmaceutical-adjacent contexts: AI-generated care plans should function as evidence-based scaffolding that clinicians customise, rather than as final prescriptions. Framing the Care Plan feature as a 'starting point for physician customisation' rather than a complete prescriptive output may improve both physician acceptance and clinical safety.

5.6 Adoption Pathway and Implementation Framework

The finding that 85.7% of respondents endorsed either an immediate or conditional pilot represents a substantial mandate for clinical evaluation. The 52.4% favouring an 'Immediate Pilot' aligns with the identification of KOLs, Key Opinion Leaders, as both a target physician profile and a pilot design element.

There is well-established precedent in digital health implementation science for KOL-anchored pilots as a mechanism for generating credible local evidence that can accelerate broader adoption [15]. In the Indian context, a structured pilot with 10–20 high-volume OPD physicians across 3–4 major therapeutic areas (chronic disease, respiratory, cardiology, and a non-specialist general medicine setting) would provide sufficient statistical power to evaluate documentation time, clinical note quality, and physician experience using validated instruments.

The concern about platform speed in 30–40% of scenarios, raised in Dr. Nivangune's written feedback, is a specific, actionable performance metric. In high-volume Indian OPDs where consultation cycles may be as short as 3–5 minutes, any AI-generated note must be available within the consultation cycle for review before the patient exits. Latency benchmarking under Indian network conditions (including 4G mobile internet, which is the most common connectivity modality in outpatient settings) should be a formal specification requirement before pilot commencement.

6. SUMMARY TABLES

Table 1. Pre-Demonstration Survey Results (n=23 respondents)

Survey Item	Leading Response	% Selecting
Documentation burden (scale 1–5)	Score 4 (high)	60.9%
Most valued AI scribing aspect	Real-time voice-to-text	39.1%
Primary adoption concern	Indian accent accuracy	65.2%
AI clinical reasoning familiarity	Moderately familiar	47.8%
AI role in clinical reasoning	Predict disease progression	39.1%
Follow-up visit challenge	All of the above (combined)	78.3%
No.1 AI success metric	Reduction in time per patient file	47.8%
Target doctor profile	High patient-footfall OPD doctors	60.9%

Table 2. Post-Demonstration Survey Results (n=21 respondents)

Survey Item	Leading Response	% Selecting
AI Scribing accuracy	Adequately / Exceptionally well	76.2%
Confidence in Clinical Reasoning	High confidence	76.2%
Care Plan generation rating	Somewhat effective	38.1%
Patient Notes / Visit Summarisation	Critical for long-term care	71.4%
User Experience	Seamless integration	61.9%
Strongest value proposition	Ambient Scribing (time-saving)	65.0%
Regional language need	Yes , all local languages	81.0%

Survey Item	Leading Response	% Selecting
Confidence in documentation quality	Very confident	42.9%
Final adoption recommendation	Immediate Pilot with KOLs	52.4%

Table 3. India-Specific Implementation Considerations and Recommended Mitigation Strategies

Challenge Domain	Specific Challenge	Recommended Mitigation
Multilingual NLP	22 scheduled + hundreds of dialects; code-switching in consultations	Train ASR on code-switched Hindi–English corpora; add incremental regional language modules
Data Privacy	DPDP Act 2023 compliance; patient consent at point of capture	Transient audio processing; Indian data residency; standardised consent form; ABDM alignment
Workflow Speed	Slow response in 30–40% of scenarios per expert feedback	Latency benchmarking on 4G; client-side edge processing for low-bandwidth environments
Care Plan Localisation	International guidelines may not reflect Indian prescribing norms	Indian pharmacopoeia integration; physician brand-preference customisation layer
Ambient Noise	Busy OPDs with multiple simultaneous voices	Directional microphone hardware; speaker diarisation tuned for Indian OPD acoustics
Regulatory Approval	No CDSCO or equivalent approval; no professional body endorsement	CDSCO SaMD pathway submission; seek endorsement from IMA / specialty bodies

7. CONCLUSIONS AND RECOMMENDATIONS

This pre–post survey study provides empirical evidence that a sample of Indian clinicians and medical affairs professionals perceive a high and unmet documentation burden in OPD settings, and that exposure to a structured demonstration of an ambient clinical AI platform produces substantial gains in confidence across all core feature dimensions. The findings support the following conclusions:

- Documentation burden is severe and almost universal among the Indian clinical population engaged by Lupin's Medical Affairs network, with 91.3% of pre-demonstration respondents rating burden at the two highest points on a 5-point scale.
- Post-demonstration acceptance of the Mendalia platform was high across all core features, with particularly strong endorsement of the AI Scribing (76.2% adequate or exceptional accuracy) and Visit Summarisation (95.2% useful or critical) features.
- Multilingual ASR is a non-negotiable implementation requirement for Indian deployment, endorsed by 81% of post-demonstration respondents, and represents the most technically demanding development priority for the platform.
- Data privacy compliance with India's DPDP Act 2023 and alignment with the Ayushman Bharat Digital Mission are institutional prerequisites that must be formally addressed before any commercial deployment.
- An immediate KOL-led pilot is recommended by 85.7% of respondents and constitutes the most appropriate next step, providing real-world performance data under Indian clinical conditions.

The recommendations arising from this study, relevant both to Mendalia AI's product development roadmap and to Lupin Limited's adoption strategy, are:

- Commission a structured multilingual performance evaluation of Mendalia's ASR system across Hindi, Bengali, Marathi, Telugu, Tamil, and Gujarati, using OPD-representative code-switched audio corpora.
- Develop and publish a DPDP-compliant patient consent framework, data architecture document, and privacy impact assessment for Indian deployment, with specific reference to SAHI and BODH mechanisms and ABDM interoperability.
- Conduct a KOL-anchored, prospective real-world pilot study across 3–4 therapeutic areas (diabetes, cardiovascular disease, respiratory medicine, and general medicine) measuring documentation time, note quality using validated rubrics, physician burnout scores, and patient satisfaction as primary endpoints.
- Engage the Indian Medical Association (IMA) and relevant specialty bodies (e.g., Research Society for the Study of Diabetes in India [RSSDI], Cardiological Society of India [CSI]) for formal platform review and endorsement consideration.
- Develop a localised Indian care plan layer that incorporates commonly used Indian generic and branded molecules, enabling AI-generated treatment plans to serve as a locally relevant starting scaffold for physician customisation.

Ambient clinical AI represents a potentially transformative technology for a healthcare system under profound structural stress. India's physician workforce is too small, too unevenly distributed, and too heavily burdened to sustain high-quality contemporaneous documentation at scale without technological assistance. The evidence presented in this study suggests that Mendalia AI's platform is technically credible, well-received by Indian clinical professionals after direct exposure, and appropriately positioned for a structured pilot. The imperative for rigorous prospective evidence, cultural adaptation, regulatory compliance, and linguistic inclusivity is equally clear. Addressing these requirements systematically will determine whether ambient AI scribing fulfils its considerable promise in the world's most complex and most consequential healthcare market.

8. LIMITATIONS

This study has several limitations that should be acknowledged. First, the sample size (n=23 pre-demonstration, n=21 post-demonstration) is small, and respondents were drawn from a single pharmaceutical company's Medical Affairs network, which may not be representative of the broader Indian clinical community. Second, the demonstration was conducted in a controlled, simulated environment using curated clinical scenarios; real-world performance in ambient noise conditions, across a broader range of regional languages, and with authentic patients may differ substantially. Third, as a pre–post design without a control group, the study cannot attribute changes in sentiment exclusively to platform exposure; order effects, social desirability bias, and selection effects cannot be fully excluded. Fourth, the survey did not capture respondents' clinical specialties or years of experience, limiting subgroup analysis. Future studies should employ larger, stratified samples, include real-world clinical observation, and use validated technology acceptance and physician burnout instruments as quantitative endpoints.

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